

Clock — Post-1, 2, 3, 4

1. On Mirror Image
2. On Water Image
3. On Degree
4. On Symbolic

→ Mirror Image → (left → Right) turns
 Right → Left

1	1
2	8
3	3
4	4
5	2
6	9
7	F
8	8
9	P
10	01
11	11
12	21

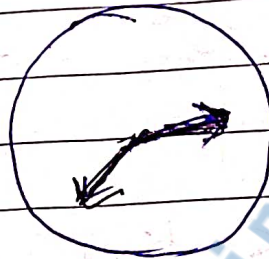
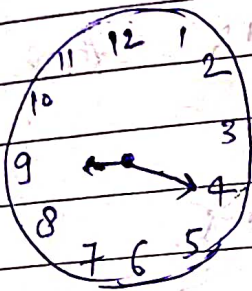
→ Water Image →

up → down
 down → up

1	2	3	4	5	6
J	5	3	4	2	e
7	8	9	10	11	12
J	8	d	J0	JJ	J5

Q → Clock time → 09:20

time of mirror image → ?



Ans - 2:40

Alternate:

$$12:00 - 9:20 = 2:40$$

Q → Clock time → 8:10

Mirror time → 11:60 - 8:10

= 3:50 Ans

Q → Clock time → 12:10

Mirror time → 11:60 - 12:10

→ 11:60 - 0:10

→ 11:50 Ans

Q → Clock time → 11:15

Mirror time → 11:60 - 11:15

→ 00:45 → 12:45

Q, Mirror time $\rightarrow 12:45$

Clock time $\rightarrow 11:60 - 12:45$

$\rightarrow 11:15$ Ans

Q, Clock time $\rightarrow 5:45$

Mirror time $\rightarrow 12:00 - 5:45$

$\rightarrow 6:15$ Ans

Q, Clock time $\rightarrow 11:55$

Mirror time $\rightarrow 11:60 - 11:55$

$\rightarrow 00:05 \rightarrow 12:05$ Ans

Q, Mirror time $\rightarrow 11:25$

Clock time $\rightarrow 11:60 - 11:25$

$\rightarrow 00:35 \rightarrow 12:35$ Ans

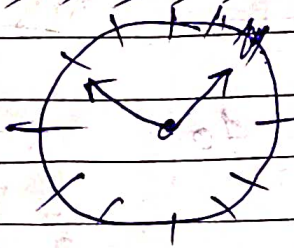
Q, Mirror time $\rightarrow 12:30$

Clock time $\rightarrow 11:60 - 12:30$

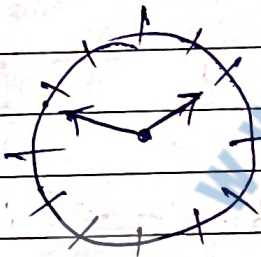
$\rightarrow 11:60 - 00:30$

$\rightarrow 11:30$ Ans

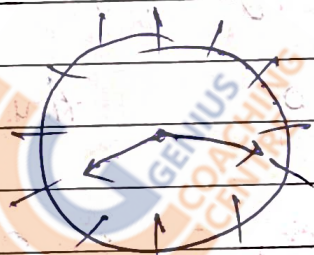
Q.

Clock time $\rightarrow$ 8:20Water Image $\rightarrow$  $\rightarrow$ 8:20 $\rightarrow$ 9:10 $\rightarrow$

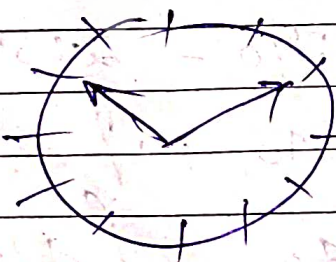
8:20

 $\rightarrow$

9:10

 $\rightarrow$

8:20

 $\rightarrow$

10:10

Printy

9:10 $\rightarrow$ (2)
10:10 $\rightarrow$ (1)

Alternate:

17:30 $\rightarrow$ 8:2017:40 $\rightarrow$ 8:20

9:10 + 1

10:10 $\leftarrow$

Q, Clock time $\rightarrow 7:25$

Water time $\rightarrow 17:30 - 7:25$

$\rightarrow 10:05 + 1$

$\rightarrow \underline{11:05}$ Ans

Q, Clock time $\rightarrow 8:40$

Water time $\rightarrow 17:90 - 8:40$

$\rightarrow 09:50 + 11$

$\rightarrow 10:50$ Ans

Q, Clock time $\rightarrow 09:20$

Water time $\rightarrow 17:30 - 09:20$

$\rightarrow 08:10 + 1$

$\rightarrow 09:10$ Ans

Q, Clock time $\rightarrow 10:00 \cancel{22:00} 2:50$

Water time $\rightarrow 17:90 - 2:50$

$\rightarrow 15:40 + 1 \quad 3:40 + 1$

$\rightarrow 16:40 \quad 4:40$ Ans

Part-3

Degree Problem

Q

Clock time $\rightarrow$ 3:40

Angle between hour hand & minute hand

1 hour (h. hand) $\rightarrow$ 30°

1 minute (m. hand) $\rightarrow$ 6°

1 minute (h. hand) $\rightarrow$ $\frac{30^\circ}{60} = 0.5^\circ$

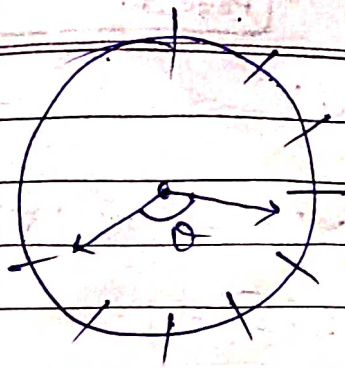
1 hour (m. hand) $\rightarrow$ 360°

Minute hand $\rightarrow$ 60 min $\rightarrow$ 360°

Hour hand $\rightarrow$ 12 hours $\rightarrow$ 360°

Minute hand $\rightarrow$ $6^\circ/\text{min}$

Hour hand $\rightarrow$ $0.5^\circ/\text{min}$



→ 3:40

$$\theta = 150^\circ - 20^\circ = 130^\circ \text{ Ans}$$

Note: Relative speed of minute hand and hour hand.

$$R.S = 6^\circ - 0.5 = 5.5^\circ/\text{min}$$

Difference between both hand is $5.5^\circ/\text{min}$ or $\frac{11}{2}^\circ/\text{min}$ and

hence difference in one hour is $\frac{11}{2}^\circ \times 60 = 33^\circ$

Clock divided into 12 blocks

$$\text{So } \left\{ \begin{array}{l} 12 \text{ blocks} = 360^\circ \\ 1 \text{ block} = 30^\circ \end{array} \right\}$$

Now 1 block = 5 sub blocks

$$\left\{ \begin{array}{l} 5 \text{ subblocks} = 30^\circ \\ 1 \text{ subblock} = 6^\circ \end{array} \right\}$$

$$\text{Angle} = |30H - \frac{11}{2}M|$$

Q₁ Clock time $\rightarrow$ 4:20

$$\begin{aligned}\text{Angle} &= |30 \times 4 - \frac{11}{2} \times 20| \\ &= |120 - 110| \\ &= 10^\circ\end{aligned}$$

Q₂ Clock time $\rightarrow$ 6:24

$$\begin{aligned}\text{Angle} &= |30 \times 6 - \frac{11}{2} \times 24| \\ &= |180 - 132| \\ &= 48^\circ\end{aligned}$$

Q₃ Clock time $\rightarrow$ 10:10

$$\begin{aligned}\text{Angle} &= |30 \times 10 - \frac{11}{2} \times 10| \\ &= |300 - 55| \\ &= 245^\circ \rightarrow 360 - 245 \\ &\rightarrow 115^\circ \text{ Ans}\end{aligned}$$

Clock

Q → Clock time → 12:05

$$\begin{aligned}
 \text{Angle} &= \left| 30 \times 12 - \frac{11}{2} \times 5 \right| \\
 &= \left| 30 \times 0 - \frac{11}{2} \times 5 \right| \\
 &= \left| \frac{11}{2} \times 5 \right| = \left| \frac{55}{2} \right| = 27.5^\circ \quad \text{Ans}
 \end{aligned}$$

Q → Clock time = 8:20

$$\begin{aligned}
 \text{Angle} &= \left| 8 \times 30 - \frac{11}{2} \times 20 \right| \\
 &= 240 - 110 \\
 &= 130^\circ \quad \text{Ans}
 \end{aligned}$$

Q → Clock time = 11:20

$$\begin{aligned}
 \text{Angle} &= \left| 11 \times 30 - 20 \times \frac{11}{2} \right| \\
 &= \left| 330 - 110 \right| \\
 &= \left| 220^\circ \right| = \left| 140^\circ \right| \quad \text{Ans}
 \end{aligned}$$

Q → Clock time = 12:40

$$\begin{aligned}
 \text{Angle} &= \left| 12 \times 30 - \frac{11}{2} \times 40 \right| \\
 &= 360 - 220 \\
 &= \left| 140^\circ \right| \quad \text{Ans}
 \end{aligned}$$

Q

Clock time 1 = 8:00

Clock time 2 = 2:00

Angle between period of hour hand

$$= 6 \times 30 = 180^\circ \text{ Ans}$$

Q

Clock time 1 = 4:00

time after 90° rotation

$$4:00 + 3 \times = 7:00$$

→ both hand parallel

→ both hand perpendicular

→ both hand opposite

→ both hand have a distance of n minute.

Q

In between 4:00 & 5:00

when both happen will parallel.

So,

$$\text{Time} = \frac{\text{Distance}}{\text{Relative speed}} = \frac{120}{11/2}$$

$$= \frac{240}{11} = 21\frac{9}{11} \text{ min} \Rightarrow \text{Ans} = 4:21\frac{9}{11} \text{ min}$$

$$\text{Time} = (30H \pm \theta) \times \frac{2}{11}$$

Alternate

$$\text{Time} = (30 \times 4 \pm \theta) \times \frac{2}{11}$$

$$= 120 \times \frac{2}{11} = \frac{240}{11} = 21 \frac{9}{11} \text{ hr}$$

Q

In between 2:00 & 3:00 when both hand will opposite →

$$\text{Time} = (30 \times 2 \pm 180) \times \frac{2}{11} = \frac{240}{11} \times 2 = \frac{480}{11} = 43 \frac{7}{11}$$

$$= (60 \pm 180) \times \frac{2}{11}$$

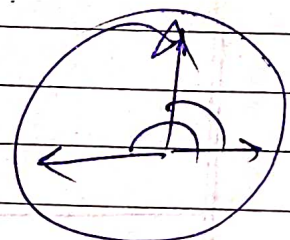
$$= 240 \times \frac{2}{11}$$

$$= \frac{480}{11} = 43 \frac{7}{11}$$

Q

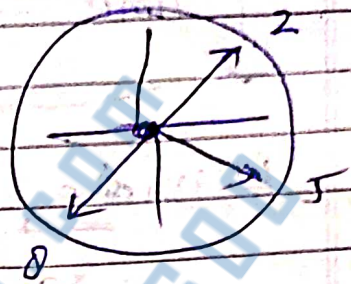
In between 9:00 & 10:00 when both hands will opposite.

$$\text{Time} = \frac{90}{1\frac{1}{2}} = \frac{180}{11} = 16 \frac{4}{11}$$



Q. in between 5 & 6 angle 90° .

$$\begin{aligned} \text{Time} &= \frac{60}{11} \times 2 \\ &= \frac{120}{11} = 10 \frac{10}{11} \end{aligned}$$



$$\text{Time} = \frac{240}{11} \times 2 = \frac{480}{11}$$

→ 1 hour →

- ① 1 time parallel
- ② 1 time opposite
- ③ 2 time perpendicular.
- ④ 2 time every except parallel and opposite.

→ 12 hour →

- ① 11 time parallel →
- ② 11 time opposite →
- ③ 22 time perpendicular

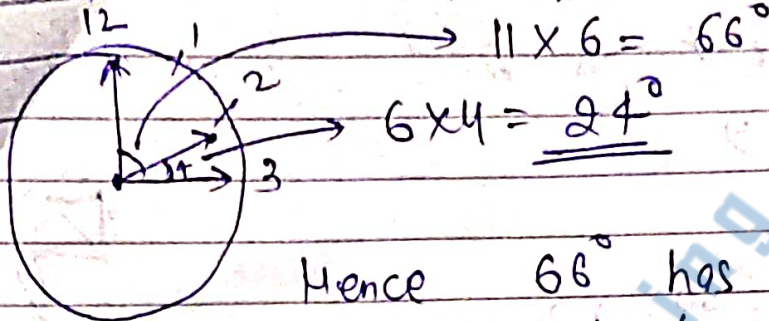
11:00 → 1:00

3:00 & 9:00

→ 24 hour →

- ① 22 time parallel
- ② 22 time opposite
- ③ 44 time perpendicular.

In between 3:00 & 4:00 →
when hour hand is leading by
4 minutes with minute hand.



Hence 66° has to travel
by minute hand for condition

$$\text{So Time} = \frac{\text{Distance}}{\text{R.S}} = \frac{66^\circ}{11/2} = \frac{66 \times 2}{11}$$

$$= \frac{132}{11} = 12 \text{ min}$$

Q, 4:56 to 5:32, time in between these two
reference, how much past of an hour is -
a) $\frac{1}{4}$ b) $\frac{3}{4}$ c) $\frac{3}{5}$ d) $\frac{1}{2}$

Q, In a clock wall, one day the time
was corrected to 8:00 AM. If the
clock becomes faster 10 minutes in 24 hours.
Now Next day at 1:00 PM what will be
the real time?

$$\left. \begin{array}{l} 8:00 \text{ AM} \rightarrow 8:00 \text{ AM} \rightarrow 10 \text{ min} \\ 1:00 \text{ PM} \rightarrow 1:00 \text{ PM} \rightarrow 5 \text{ hours} \rightarrow \frac{10}{24} \times 5 = \frac{50}{24} \end{array} \right\}$$

Date : / /

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If both the hour hand and minute hand move at their normal speeds, then both the hands meet after how many minutes?

$$\text{Time} = \frac{360}{11} \times 2$$

$$= \frac{720}{11} = 65\frac{5}{11}$$



$$\text{Time} = (30H \pm 0) \times \frac{2}{11}$$

$$= (30 \times 12 \pm 0) \times \frac{2}{11}$$

$$360 \times \frac{2}{11} = \frac{720}{11}$$

$$= 65\frac{5}{11}$$